

An Introduction to R

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Ce cours a été construit par Céline Engelbeen



- 1 Basics
- 2 Working with Scripts
- 3 Data Types and Data Structures
- 4 Statistics et probabilités
- 5 Managing, Reading and Writing Data
- 6 Selecting Data and Variables
- 7 Managing Environment
- 8 Statistics and Plotting for Multidimensional Data



1 Basics

2 Working with Scripts

3 Data Types and Data Structures

4 Statistics et probabilités

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6 Selecting Data and Variables

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8 Statistics and Plotting for Multidimensional Data



Some of these slides are freely inspired by :

- "An Interactive Introduction to R for Actuaries", by Driscoll and Murphy, CAS Conference, November 2009, Boston, Massachusetts. <https://www.casact.org/education/annual/2009/handouts/c11-driscoll.pdf>
- "Introduction aux tests statistiques avec R" by Christophe Chesneau, Licence, France, 2016, cel-0.1387707v2.
- "R introduction and reference guide", School of Mathematics and Statistics, University of Melbourne.



- Data Manipulation
 - Connecting to data sources
 - Slicing and selecting data
- Modeling and Computation
 - Statistical modeling
 - Data Mining
 - Machine Learning
 - Numerical simulation
- Data Visualization
 - Visualizing fit of models
 - Composing statistical graphics



R GUI (64-bit)
Fichier Fenêtres Edition Aide

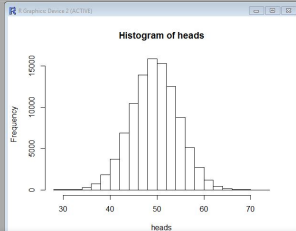
```
R Console
Tapez 'license()' ou 'licence()' pour plus de détails.
R est un projet collaboratif avec de nombreux contributeurs.
Tapez 'contributors()' pour plus d'information et
'citation()' pour la façon de le citer dans les publications.
Tapez 'demo()' pour des démonstrations, 'help()' pour l'aide
en ligne ou 'help.start()' pour obtenir l'aide au format HTML.
Tapez 'q()' pour quitter R.

[Sauvegarde de la session précédente restaurée]

> weight <- c(110, 180, 240)
> height <- c(5.5, 6.1, 6.2)
> bmi <- (weight*4.05)/height^2
> bmi
[1] 17.74545 23.60656 30.46826
> plot(weight, height)
> heads <- rbinom(10^5, 100, 0.50)
> hist(heads)
> TRUE
starting httpd help server ... done
> mydata<-rbind(weight, height)
> fix(mydata)
```

R Éditeur de données

	row.names	col1	col2	col3
1	weight	110	180	240
2	height	5.5	6.1	6.2
3				
4				
5				



R: Histograms

127.0.0.1:16839/library/graphics/html/hist.html

hist {graphics}

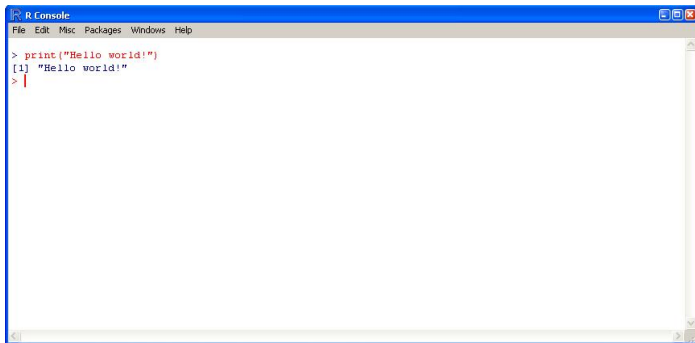
R Documentation

Histograms

Description

The generic function `hist` computes a histogram of the given data values. If `plot = TRUE`, the resulting object of class "histogram" is plotted by `plot.histogram`, before it is returned.

Usage



```
R Console
File Edit Misc Packages Windows Help

> print("Hello world!")
[1] "Hello world!"
> |
```

You said austere? Yep, but it's just a beginning...

RStudio is an IDE (Integrated Development Environment) which facilitate working with R.

RStudio console mode

The screenshot shows the RStudio interface with the console, environment, and plots panes. The console pane contains the following commands and output:

```
[3,] 3 0 1
> head(A, n=2)
      [,1] [,2] [,3]
[1,]  1  0 -1
[2,]  2  1  0
> summary(A)
      V1      V2
Min.   :1.0   Min.   :0.0000
1st Qu.:1.5   1st Qu.:0.0000
Median :2.0   Median :0.0000
Mean   :2.0   Mean   :0.3333
3rd Qu.:2.5   3rd Qu.:0.5000
Max.   :3.0   Max.   :1.0000
      V3
Min.   :-1.0
1st Qu.: -0.5
Median : 0.0
Mean   : 0.0
3rd Qu.: 0.5
Max.   : 1.0
> rm(B)
> ls()
[1] "A"
> x<-rnorm(1000,3,2)
> hist(x)
> setwd("~/c:/")
> |
```

Annotations in the image:

- A blue box highlights the console pane, with an arrow pointing to it and the text: (1) Type your commands here.
- An orange box highlights the Environment pane, with an arrow pointing to it and the text: (2) See the results here and/or in « Environment » and/or « Plots ».

The Environment pane shows the following data:

Variable	Class	Summary
A	num	[1:3, 1:3] 1 2 3 ...
x	num	[1:1000] 1.302 2.20...

The Plots pane shows a histogram titled "Histogram of x" with the x-axis labeled "x" and the y-axis labeled "Frequency". The histogram shows a distribution of values between -2 and 10, with a peak around 4.

- Simple Math

```
2+2
```

```
[1] 4
```

```
log(100)
```

```
[1] 4.60517
```

- Storing Results in Variables, using '<-' operator

```
x<-2+2
```

```
(x*3 +1) / 2
```

```
[1] 6.5
```

```
sqrt(121)
```

```
[1] 11
```



Help within R itself for a function

```
help(func)
```

```
?func
```

For a topic

```
help.search("binomial")
```

```
## starting httpd help server ... ## done
```

```
??poisson
```



- Repeat the same number several times:

```
rep(3, 10)
```

```
[1] 3 3 3 3 3 3 3 3 3 3
```

- Sequences of integers:

```
4:9
```

```
[1] 4 5 6 7 8 9
```

```
10:-4
```

```
[1] 10 9 8 7 6 5 4 3 2 1 0 -1 -2 -3 -4
```

- A sequence from a starting number to and ending number by a chosen step:

```
seq(0.25, 0.80, by=0.05)
```

```
[1] 0.25 0.30 0.35 0.40 0.45 0.50 0.55 0.60 0.65 0.70 0.75 0.80
```

- A sequence from a starting number to and ending number by with a chosen length:

```
seq(0, 1, length.out = 9)
```

```
[1] 0.000 0.125 0.250 0.375 0.500 0.625 0.750 0.875 1.000
```

- Choose a sample: choose randomly 5 numbers between 1 and 100 without repetition:

```
sample(1:100, 5)
```

```
[1] 60 39 18 49 20
```

- Choose a sample: choose 15 numbers between 1 and 3 with repetition:

```
sample(1:3, 15, replace = TRUE)
```

```
[1] 1 1 3 3 3 2 3 2 3 1 1 2 3 2 2
```



- If you want to create simulations that can be reproduced :

```
set.seed(3)  
sample(1:100, 5)
```

```
[1] 5 58 12 36 99
```

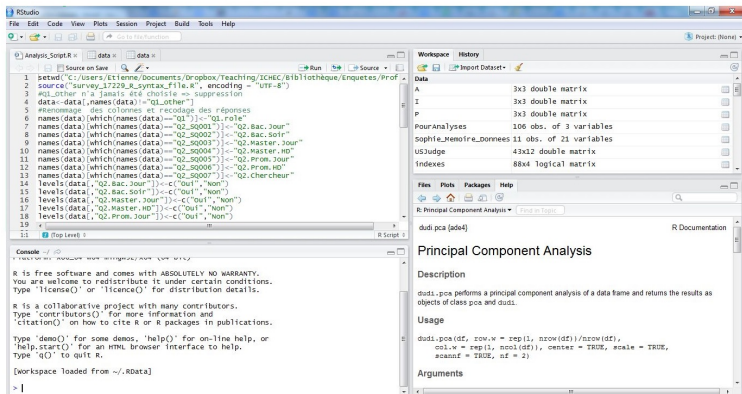
```
set.seed(3)  
sample(1:100, 5)
```

```
[1] 5 58 12 36 99
```

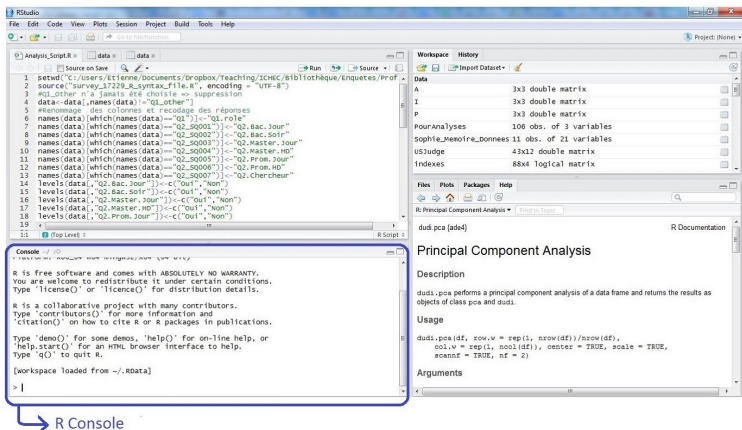
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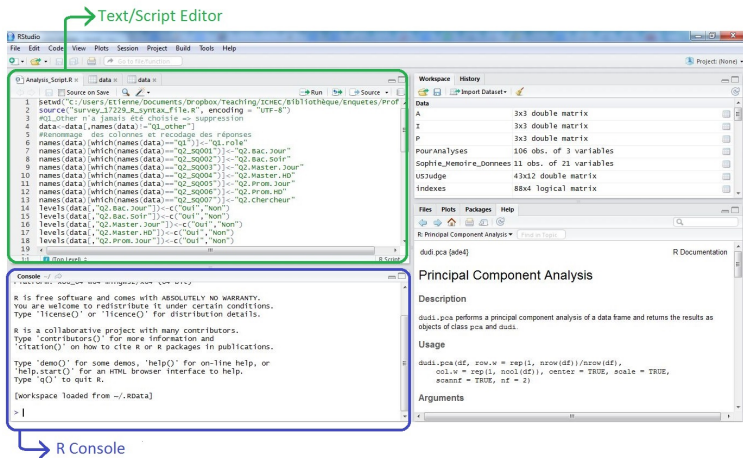
IDE = Integrated Development Environment



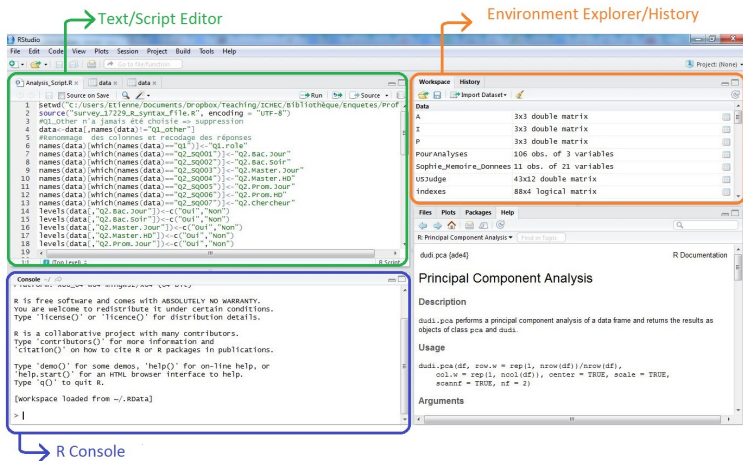
IDE = Integrated Development Environment



IDE = Integrated Development Environment



IDE = Integrated Development Environment

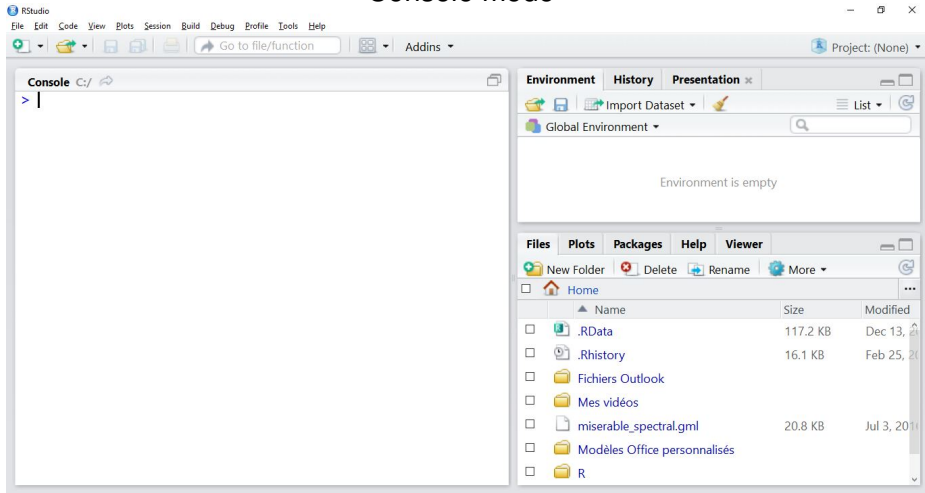


IDE = Integrated Development Environment

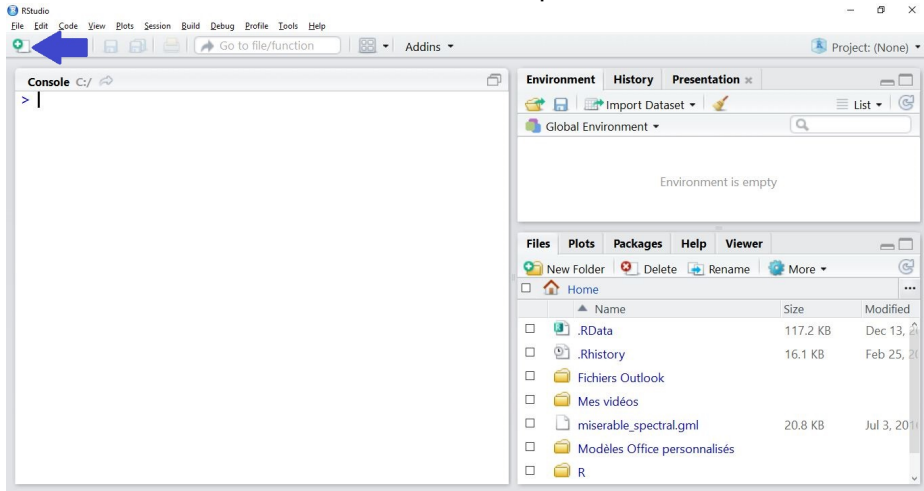
The screenshot shows the RStudio IDE interface with four main panels and their functions indicated by arrows:

- Text/Script Editor:** The top-left panel where R code is written. It contains a script named 'Analysis_Script.R' with R code for loading data and performing a Principal Component Analysis (PCA).
- Environment Explorer/History:** The top-right panel showing the current workspace. It lists objects: 'A' (3x3 double matrix), 'I' (3x3 double matrix), 'P' (3x3 double matrix), 'PourAnalyses' (106 obs. of 3 variables), 'Sophie_Memoire_Donnees' (11 obs. of 21 variables), 'USJudge' (43x12 double matrix), and 'indexes' (88x4 logical matrix).
- R Console:** The bottom-left panel showing the R prompt and output. It displays the R welcome message and the status '(workspace loaded from ~/Rdata)'. The prompt is '> |'.
- Help, Packages Loader, Plots:** The bottom-right panel showing the 'R: Principal Component Analysis' help page. It includes a description of the 'dudi.pca' function, its usage, and arguments.

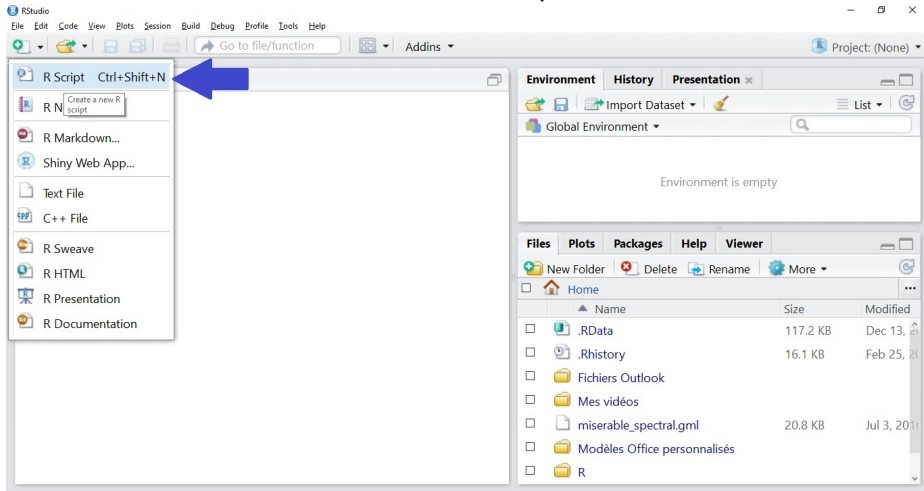
Console mode



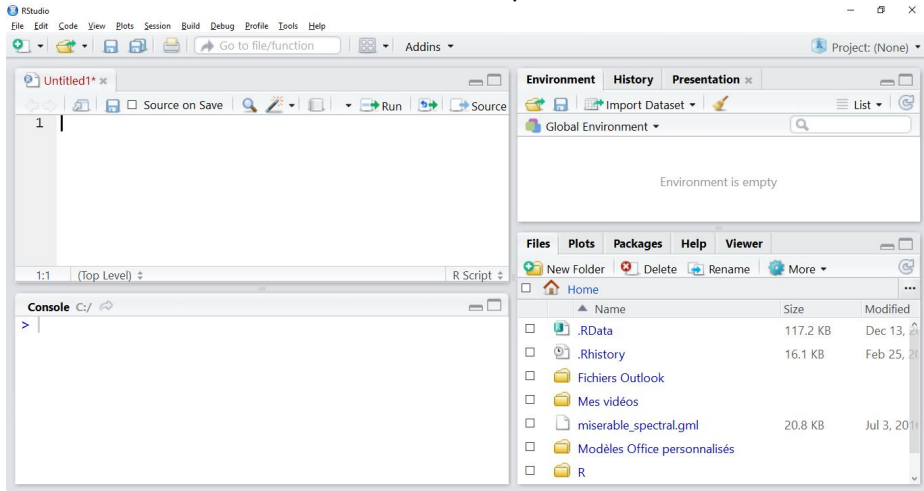
Create a new script file



Create a new script file



Your new script file



Type your code

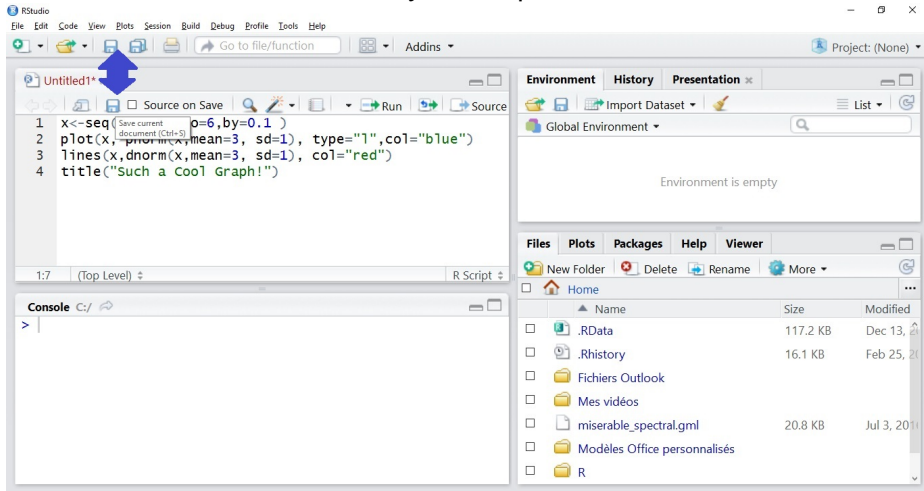
The screenshot displays the RStudio IDE interface. The main editor window, titled 'Untitled1*', contains the following R code:

```
1 x<-seq(from=0, to=6,by=0.1 )
2 plot(x, pnorm(x,mean=3, sd=1), type="l",col="blue")
3 lines(x,dnorm(x,mean=3, sd=1), col="red")
4 title("Such a Cool Graph!")
```

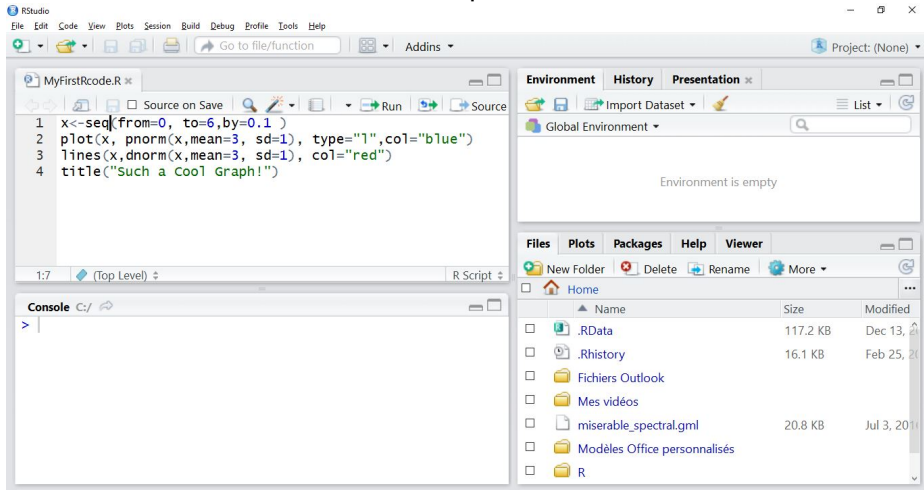
The console at the bottom left shows the prompt '> |'. The environment pane on the right indicates 'Environment is empty'. The file explorer on the bottom right shows the 'Home' directory with the following files and folders:

Name	Size	Modified
.RData	117.2 KB	Dec 13, 2016
.Rhistory	16.1 KB	Feb 25, 2017
Fichiers Outlook		
Mes vidéos		
miserable_spectral.gml	20.8 KB	Jul 3, 2016
Modèles Office personnalisés		
R		

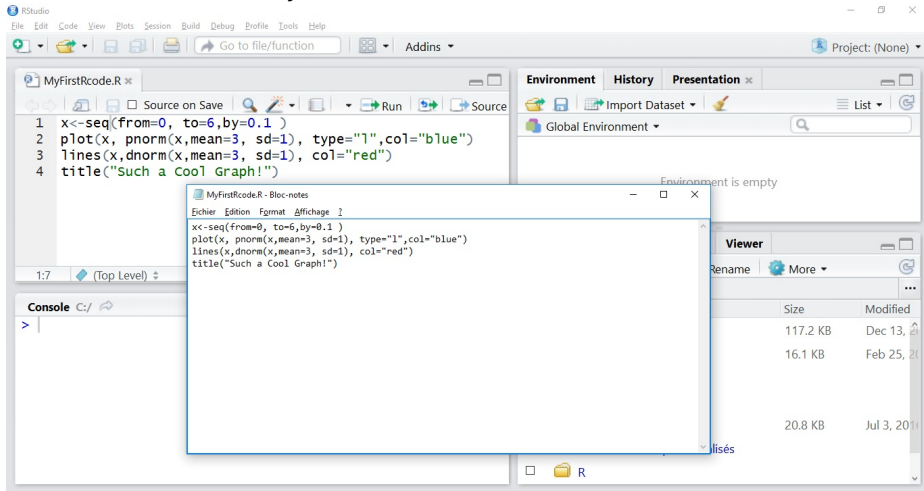
Save your script file



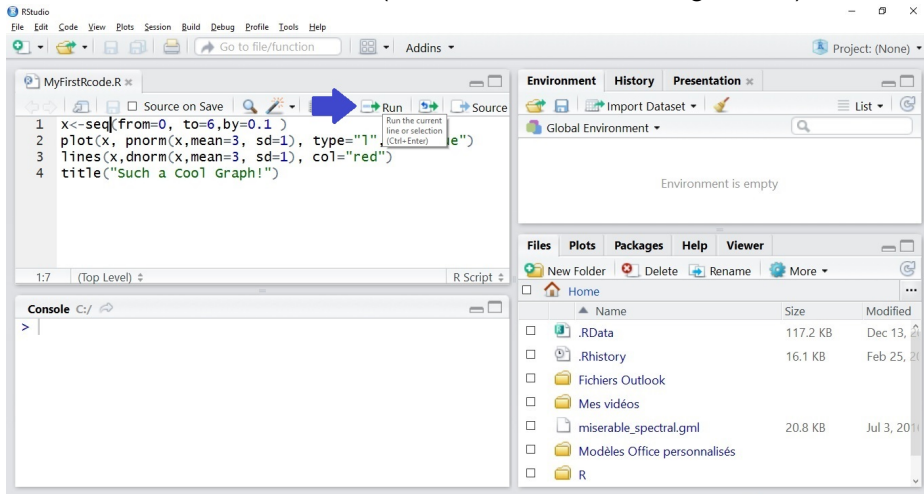
Your script file...



...is just a text file with extension .R



Run the current line (line where is the blinking cursor)



Run the current line (done)

The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains a script named `MyFirstRcode.R` with the following R code:

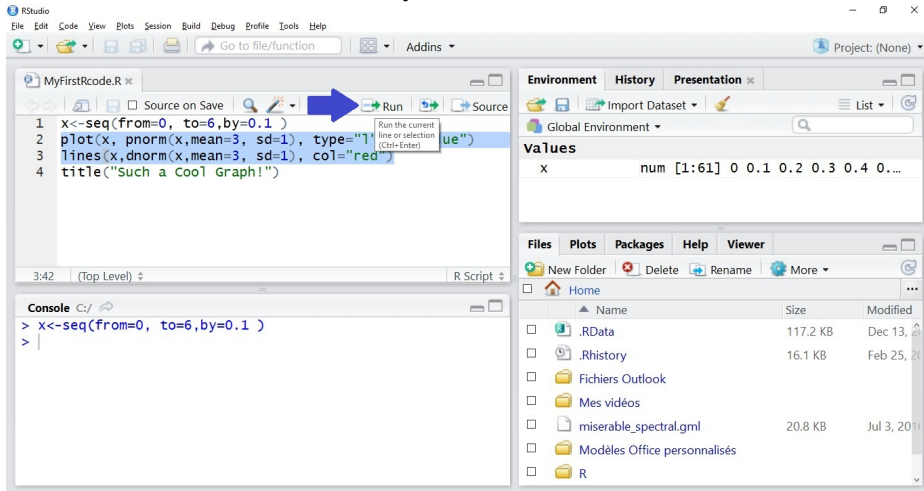
```
1 x<-seq(from=0, to=6,by=0.1 )
2 plot(x, pnorm(x,mean=3, sd=1), type="l",col="blue")
3 lines(x,dnorm(x,mean=3, sd=1), col="red")
4 title("Such a Cool Graph!")
```
- Console:** Shows the execution of the first line of code:

```
> x<-seq(from=0, to=6,by=0.1 )
> |
```
- Environment Panel:** Displays the `Global Environment` with a table of values for the variable `x`.

num	[1:61]	0	0.1	0.2	0.3	0.4	0...
x							
- Files Panel:** Shows the file explorer with a list of files and folders in the `Home` directory.

Name	Size	Modified
.RData	117.2 KB	Dec 13, 2016
.Rhistory	16.1 KB	Feb 25, 2017
Fichiers Outlook		
Mes vidéos		
miserable_spectral.gml	20.8 KB	Jul 3, 2016
Modèles Office personnalisés		
R		

Run your selection



The screenshot shows the RStudio interface with a script named 'MyFirstRcode.R' open. The script contains the following R code:

```
1 x<-seq(from=0, to=6,by=0.1 )
2 plot(x, pnorm(x,mean=3, sd=1), type="l", col="blue")
3 lines(x,dnorm(x,mean=3, sd=1), col="red")
4 title("Such a Cool Graph!")
```

A blue arrow points to the 'Run' button in the toolbar. A tooltip over the 'Run' button reads: 'Run the current line or selection (Ctrl+Enter)'. The console shows the execution of the script:

```
> x<-seq(from=0, to=6,by=0.1 )
> |
```

The 'Environment' pane on the right shows the 'Global Environment' with a variable 'x' of type 'num' [1:61] containing values from 0 to 0.6.

The 'Files' pane on the right shows a file explorer view of the 'Home' directory, listing files such as '.RData', '.Rhistory', 'Fichiers Outlook', 'Mes vidéos', 'miserable_spectral.gml', 'Modèles Office personnalisés', and 'R'.

Run your selection (done)

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Project: (None)

MyFirstRcode.R

```
1 x<-seq(from=0, to=6,by=0.1)
2 plot(x, pnorm(x,mean=3, sd=1), type="l",col="blue")
3 lines(x,dnorm(x,mean=3, sd=1), col="red")
4 title("Such a Cool Graph!")
```

3:42 (Top Level) R Script

Console

```
> x<-seq(from=0, to=6,by=0.1)
> plot(x, pnorm(x,mean=3, sd=1), type="l",col="blue")
> lines(x,dnorm(x,mean=3, sd=1), col="red")
> |
```

Environment History Presentation

Global Environment

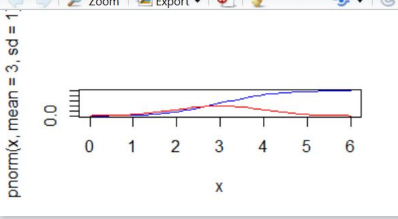
Values

x	num [1:61] 0 0.1 0.2 0.3 0.4 0...
---	-----------------------------------

Files Plots Packages Help Viewer

Zoom Export

pnorm(x, mean = 3, sd = 1)



Run the whole file

The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains the script `MyFirstRcode.R` with the following code:

```
1 x<-seq(from=0, to=6,by=0.1 )
2 plot(x, pnorm(x,mean=3, sd=1), type="l",col="blue")
3 lines(x,dnorm(x,mean=3, sd=1), col="red")
4 title("Such a Cool Graph!")
```

A blue arrow points to the `Source` button in the toolbar.
- Environment:** Shows the `Global Environment` with a variable `x` of type `num` [1:61].
- Console:** Shows the execution of the script:

```
> x<-seq(from=0, to=6,by=0.1 )
> plot(x, pnorm(x,mean=3, sd=1), type="l",col="blue")
> lines(x,dnorm(x,mean=3, sd=1), col="red")
> |
```
- Plots:** Displays a plot of the standard normal distribution. The x-axis is labeled `x` and ranges from 0 to 6. The y-axis is labeled `pnorm(x, mean = 3, sd = 1)` and ranges from 0.0 to 1.0. The plot shows a blue line for the cumulative distribution function (CDF) and a red line for the probability density function (PDF).

Run the whole file (done)

The screenshot displays the RStudio interface with the following components:

- Source Editor:** Contains the script `MyFirstRcode.R` with the following code:

```
1 x<-seq(from=0, to=6,by=0.1)
2 plot(x, pnorm(x,mean=3, sd=1), type="l",col="blue")
3 lines(x,dnorm(x,mean=3, sd=1), col="red")
4 title("Such a Cool Graph!")
```
- Console:** Shows the execution of the script:

```
> x<-seq(from=0, to=6,by=0.1)
> plot(x, pnorm(x,mean=3, sd=1), type="l",col="blue")
> lines(x,dnorm(x,mean=3, sd=1), col="red")
> source('C:/Users/EC/Dropbox/Teaching/ICHEC/MQ/Cours/MQ1-02/MyFirstRcode.R')
>
```
- Environment:** Shows the `Global Environment` with a variable `x` of type `num` [1:61].
- Plots:** Displays a plot titled `Such a Cool Graph!`. The y-axis is labeled `pnorm(x, mean = 3, sd = 1)` and the x-axis is labeled `x`. The plot shows a blue line representing the cumulative distribution function (pnorm) and a red line representing the probability density function (dnorm).

Scripts, why?

- Store your exploratory work.
- Keep a ready to use analysis.
- Automate your work.
- It's cool!



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- numeric :

```
1, 0, -1, -3.789, 2+3, ...
```

- character :

```
'O', "Hey", "This is crazy", ...
```

- logical :

```
TRUE, FALSE, 1, 0, 2==3, 2<3, ...
```

- numeric :

```
x <- c(0, 2:4)
```

```
x
```

```
[1] 0 2 3 4
```

- character :

```
y <- c("alpha", "b", "c3", "4")
```

```
y
```

```
[1] "alpha" "b" "c3" "4"
```

- logical :

```
z <- c(TRUE, FALSE, 2<=3, 2==3)  
z
```

```
[1] TRUE FALSE TRUE FALSE
```

```
z2 <- c(TRUE, FALSE, 1, 0)  
z2
```

```
[1] 1 0 1 0
```

```
● M<-matrix(c(1,4,7,2,5,8,3,6,9), ncol=3)
```

M

1	2	3
4	5	6
7	8	9

```
● S<-matrix(1:12, nrow=3, byrow=TRUE)
```

S

1	2	3	4
5	6	7	8
9	10	11	12

```
df <- data.frame(x, y, z)  
df
```

x y z 1 0 alpha TRUE 2 2 b FALSE 3 3 c3 TRUE 4 4 4 FALSE

x	y	z
0	alpha	TRUE
2	b	FALSE
3	c3	TRUE
4	4	FALSE

```
lst <- list(a=x, b=y, c=z, M=S )
```

```
lst
```

\$a

0 2 3 4

\$b

"alpha" "b" "c3" "4"

\$c

TRUE FALSE

\$M

1 2 3 4
5 6 7 8
9 10 11 12

	Linear	Tabular
Homogeneous	Vectors	Matrices
Heterogeneous	Lists	Data Frames



- Let two vectors:

```
u <- c(1:9)
v <- c(1, 0, 1, 1, 1, 0, 0, 0, 1)
```

- Operations :

```
u+5
u+v
u*v
u/v
u^2
```

(!) this does not correspond to the usual definitions of vector calculus (pay attention to the sizes of the vectors)

- Summing all entries of a vector...

```
k <- sum(u)
```

- ... or only a part of it :

```
k1 <- sum(u[1:3])
```

- Construct a vector

```
weight <- c(110, 180, 240)
```

```
height <- c(5.5, 6.1, 6.2)
```

- Vectorized Math

```
bmi <- (weight*4.88)/(height*height)
```

```
[1] 17.74545 23.60656 30.46826
```



- Let two 3x3 matrices:

```
A<-matrix(1:9, ncol=3)  
B<-matrix(c(1,0,1,1,1,0,0,0,1), nrow=3, byrow=TRUE)
```

- Vectorized operations:

```
A*2  
B+1  
A+B
```

- Matricial multiplication:

```
C<-A%*%B
```

- Determinant:

```
det(C)
```

- Inverse:

```
solve(B)
```

- Solve the system with matrice B and $b = (1, 2, 3)^t$:

$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

```
solve(B, c(1, 2, 3))
```

```
[1] -2 4 3
```



- Transpose the B :

$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^t = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$t(B)$

- Find the eigenvalues and the eigenvectors for B :

$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot X = \lambda \cdot X$$

`eigen` (B)



Basic Statistics

- Let

```
weight <- c(110, 180, 240)
```

- Mean

```
mean(weight)
```

```
[1] 176.6667
```

- Standart Deviation

```
sd(weight)
```

```
[1] 65.06407
```

- Idem

```
sqrt(var(weight))
```

```
[1] 65.06407
```

- Probability that a coin tossed 100 times gives at most 40 tails

```
pbinom(40, 100, 0.5)
```

```
[1] 0.02844397
```

- Probability that among 23 people, two share a birthday

```
pbirthday(23, 365, coincident=2)
```

```
[1] 0.5072972
```



General Principle/Schema

- `ddist()` = density function (pdf, f)
- `pdist()` = cumulative density function (cdf, F)
- `qdist()` = quantile function
- `rdist()` = random generation

"dist" must be replaced by the acronym for the wanted family of distribution

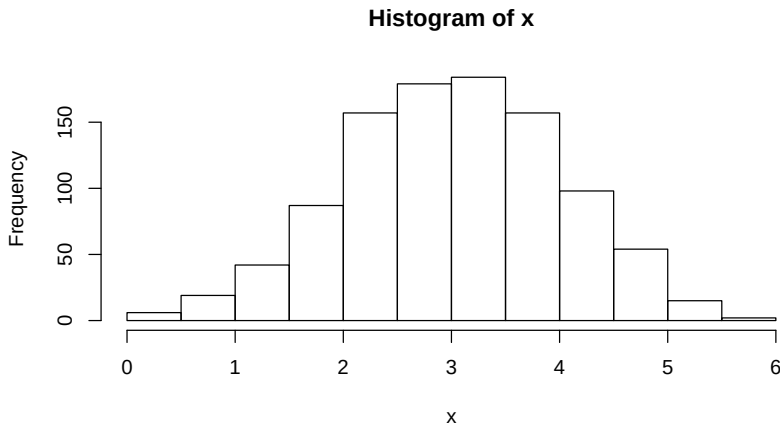
- Normal: `dnorm`, `pnorm`, `qnorm`, `rnorm`
- Binomial: `dbinom`, `pbinom`, ...
- Poisson: `dpois`, ...



Histogram of Values

```
• x <- rnorm(1000, mean=3, sd=1)
```

```
• hist(x)
```



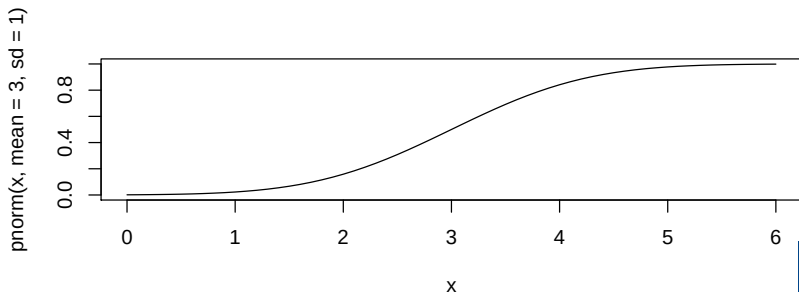
Graph for $F_{N(3,1)}(x)$

- Generating x values from 0 to 6, by step ou 0.1:

```
x<-seq(from=0, to=6, by=0.1 )
```

- Plotting the function:

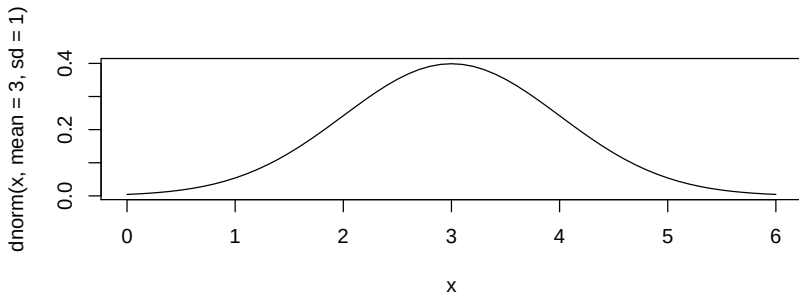
```
plot(x, pnorm(x, mean=3, sd=1), type="l")
```



Graph for $f_{N(3,1)}(x)$

- Plotting the function:

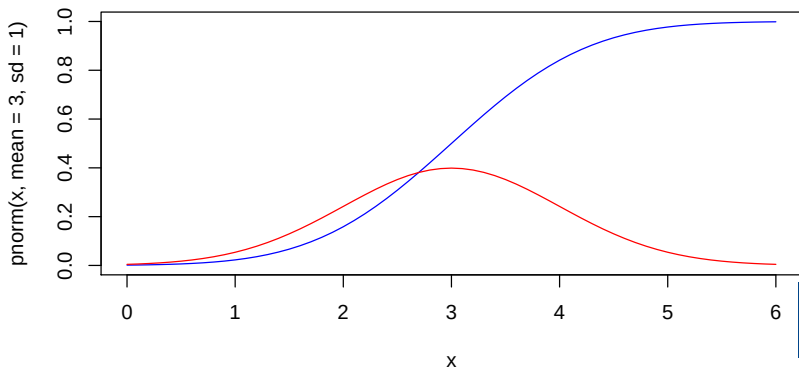
```
plot(x, dnorm(x, mean=3, sd=1), type="l")
```



Graph for $f_{N(3,1)}(x)$ and $F_{N(3,1)}(x)$ in two colors:

- Plotting the function:

```
plot(x, pnorm(x, mean=3, sd=1), type="l", col="blue")  
lines(x, dnorm(x, mean=3, sd=1), col="red")
```



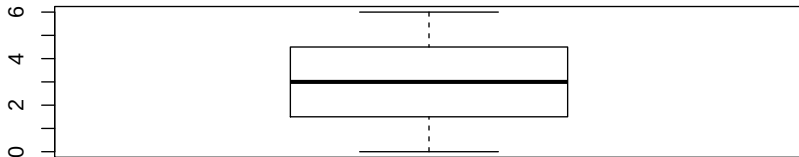
```
median(x)
```

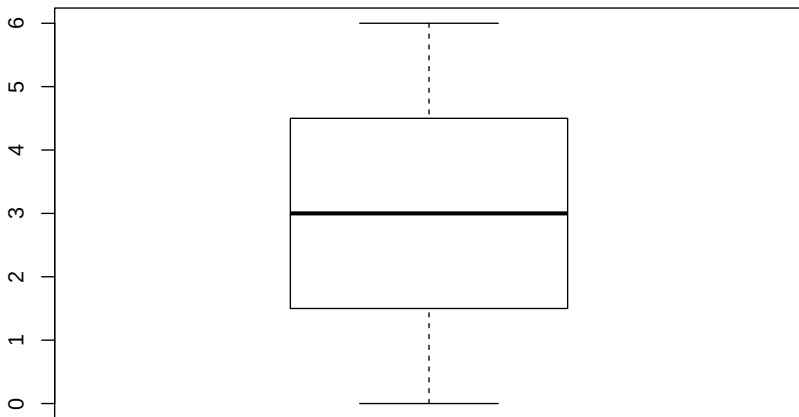
```
[1] 3
```

```
quantile(x, probs = c(0.25, 0.50, 0.75))
```

```
1.5 3.0 4.5
```

```
boxplot(x)
```





- Let the result of poll on the color of eyes

```
hair<-c("black", "black", "black", "brown",  
"brown", "brown", "brown", "blue",  
"blue", "blue", "blue", "blue",  
"blue", "blue", "green", "green",  
"green", "green")
```

- Counting

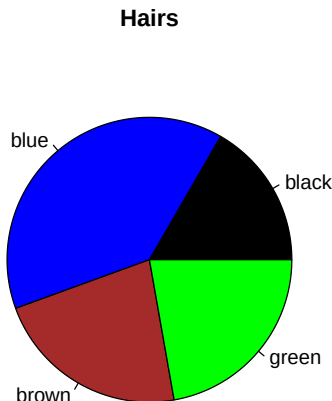
```
table(hair)
```

```
[1] "black" "blue" "brown" "green"  
3 7 4 4
```



● Plotting

```
pie(table(hair), col=c("black", "blue", "brown", "green"),  
main="Hairs")
```



- 1 Basics
- 2 Working with Scripts
- 3 Data Types and Data Structures
- 4 Statistics et probabilités
- 5 Managing, Reading and Writing Data
- 6 Selecting Data and Variables
- 7 Managing Environment
- 8 Statistics and Plotting for Multidimensional Data



- Examining a variable 'x'

```
str(x) #structure  
head(x) #first elements  
tail(x) #last elements  
class(x) #class' variable  
summary(x) #summary for the variable
```

- Specialy for Vectors and Lists

```
length(x)  
names(x) #names for the values
```

- Specialy for Data Sets and Matrices

```
View(df) #View in a dedicated window  
dim(M) #dimensions  
colnames(df) #columns names  
rownames(df) #rows names
```

- Editing a value

```
x
```

```
[1] 0 2 3 4
```

```
x[2] <- 5
```

```
x
```

```
[1] 0 5 3 4
```

- Editing names

```
names(x) <- c("1st", "2d", "3rd", "4th")
```

```
x
```

1st	2d	3rd	4th
0	5	3	4

- Editing a value

S

1	2	3	4
5	6	7	8
9	10	11	12

```
S[2,3] <- 10
```

S

1	2	3	4
5	6	10	8
9	10	11	12

- Editing a line

```
S[3, ] <- c(4, 3, 2, 1)
```

S

1	2	3	4
5	6	10	8
4	3	2	1

- Editing a column

```
S[, 3] <- S[, 3] * 2 + 1  
S
```

1	2	7	4
5	6	21	8
4	3	5	1

- Editing with table editor

```
edit(S) #Edition in a dedicated window
```

- Editing row names

```
rownames(S) <- c("L1", "L2", "L3")
```

S

L1	1	2	7	4
L2	5	6	21	8
L3	4	3	5	1

- Editing column names

```
colnames(S) <- c("a", "b", "c", "d")
```

S

	a	b	c	d
L1	1	2	7	4
L2	5	6	21	8
L3	4	3	5	1

The "Working Directory" is the default directory where R try to read or write data when you ask it.

- `getwd()`

```
[1] "C:/Users/EC/Dropbox/Teaching/ICHEC/MQ/Cours/MQ1-01"1
```

- `setwd("C:/Users/Etu/Documents/")`

- `getwd()`

```
[1] "C:/Users/Etu/Documents/"
```

¹This is the directory on my personal computer for this lesson. Your working directory could be different, for sure...

- from R environment and packages

```
data (USArrests)
```

- from R objects

```
load ("Insurance.RData")
```

- from Files

```
Insurance <- read.csv ("Insurance.csv", header=TRUE)
```

- from the Web

```
conn<-url ('http://labs.dataspora.com/test.txt')  
Insurance <- read.csv (connection, header=TRUE)
```

- from Databases

```
con<-dbConnect (driver, user, password, host, dbname)  
Insurance<-dbSendQuery (con, "SELECT * FROM claims")
```

- to R objects

```
save(Insurance, file="Insurance.RData")
```

- to Files

```
write.csv(Insurance, file="Insurance.csv")
```

- to Databases

```
con<-dbConnect(driver,user,password,host,dbname)  
dbWriteTable(con, "Insurance", Insurance)
```



- CSV Files

```
Insurance <- read.csv("Insurance.csv", header=TRUE)
```

- CSV Files (EU)

```
Insurance <- read.csv2("Insurance.csv", header=FALSE)
```

- Delimited Files (space, tab,...)

```
Insurance <- read.delim("Insurance.txt")
```

- Delimited Files (EU)

```
Insurance <- read.delim2("Insurance.dat")
```

- For Specific Formated Files

```
Insurance <- read.table("Insurance", header=FALSE,  
sep = " ", dec = ".", skip = 1)
```



- 1 Basics
- 2 Working with Scripts
- 3 Data Types and Data Structures
- 4 Statistics et probabilités
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- 6 Selecting Data and Variables
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- 8 Statistics and Plotting for Multidimensional Data



Let the following vector:

```
fruit<-c(5, 10, 1, 20)
names(fruit)<-c("orange", "banana", "apple", "peach")
```

How to select

- the 1st element?

```
fruit[1]
```

- the 1st and the 4th elements? Using a vector of positive integers:

```
fruit[c(1,4)]
```

- the three 1st elements? Using a vector of positive integers:

```
fruit[1:3]
```

- all elements except the 1st and the 3rd ? Using a vector of negative integers:

```
fruit[-c(1,3)]
```



How to select

- orange and peach values? Using a vector of character strings:

```
fruit[c("orange", "peach")]
```

- values greater than 8? Using a logical index:

```
index<-fruit>8  
fruit[index]
```

or

```
fruit[fruit>8]
```



Using the USArrest dataset: How to select

- the element at the 5th line and the 2d column:

```
USArrests[5,2]
```

- rows from 5 to 15:

```
USArrests[5:15, ]
```

nothing for the columns = no constraints on columns = all columns

- the three 1st columns:

```
USArrests[ ,1:3]
```

nothing for the rows = no constraints on rows = all rows

- elements of the ten 1st lines with columns 1 to 3:

```
USArrests[1:10,1:3]
```



How to select

- the Arizona's line:

```
USArrests["Arizona", ]
```

- the lines about Alabama, Alaska and Arizona:

```
USArrests[c("Alabama", "Alaska", "Arizona"), ]
```

- the column about murders:

```
USArrests[, "Murder"]
```

- the columns about murders and assaults:

```
USArrests[, c("Murder", "Assault")]
```

- the columns about murders and assaults for Alaska and Arizona:

```
USArrests[c("Alaska", "Arizona"), c("Murder", "Assault")]
```



- States with less than 60 assaults :

```
indexAL60<-USArrests[, "Assault"]<=60
USArrests[indexAL60, ]
```

or

```
USArrests[USArrests[, "Assault"] <= 60, ]
```

- States with urban population between 40 and 60:

```
indexUPL40<-USArrests[, "UrbanPop"]>=40
indexUPG60<-USArrests[, "UrbanPop"]<=60
betweenIndex<- indexUPL40 & indexUPG60
USArrests[betweenIndex, ]
```

where $\&$ is the logical AND operation.



How to select

- States with murder rate less than 1 or greater than 15:

```
indexML1<-USArrests[, "Murder"] <=1  
indexMG10<-USArrests[, "Murder"] >=15  
resultIndex<- indexML1 | indexMG10  
USArrests[resultIndex, ]
```

where | is the logical OR operation.

- a sample of 10 states:

```
USArrests[sample(1:50, 10), ]
```

or

```
USArrests[sample(1:dim(USArrests)[1], 10), ]
```



- 1 Basics
- 2 Working with Scripts
- 3 Data Types and Data Structures
- 4 Statistics et probabilités
- 5 Managing, Reading and Writing Data
- 6 Selecting Data and Variables
- 7 Managing Environment
- 8 Statistics and Plotting for Multidimensional Data



- Listing all objects in the environment:

```
ls()
```

- Removing an object:

```
rm(x) #removing x  
rm(list=ls()) #removing all variables
```



The screenshot shows the RStudio interface. The Environment pane on the right lists the following objects:

Object	Description
data	50 obs. of 4 variables
gap_stat	List of 6
km.res	List of 9
nb	List of 4
res.nb	List of 4
USArrests	50 obs. of 4 variables

The console output shows the results of the clustering analysis:

```

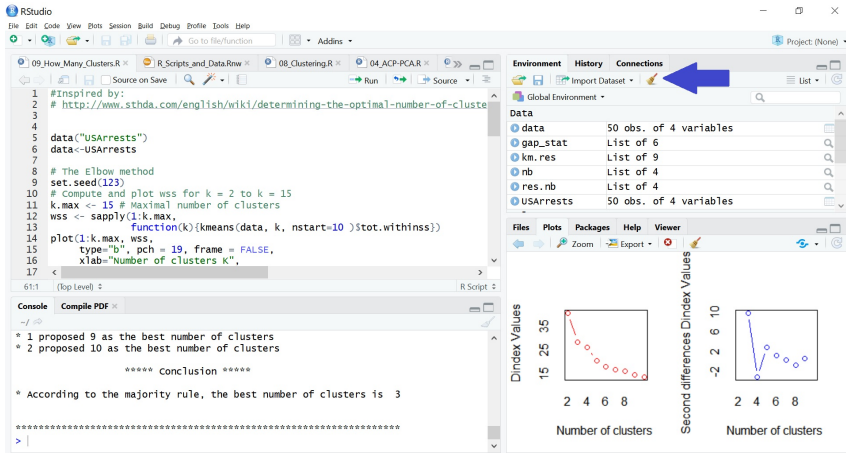
* 1 proposed 9 as the best number of clusters
* 2 proposed 10 as the best number of clusters

***** Conclusion *****

* According to the majority rule, the best number of clusters is 3
  
```

File format is .RData

Before



RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function

Addins

Project: (None)

09_How_Many_Clusters.R 08_Clustering.R 04_ACP-PCA.R

```

1 #Inspired by:
2 # http://www.sthda.com/english/wiki/determining-the-optimal-number-of-clusters
3
4
5 data("USArrests")
6 data<-USArrests
7
8 # The Elbow method
9 set.seed(123)
10 # Compute and plot wss for k = 2 to k = 15
11 k.max <- 15 # Maximal number of clusters
12 wss <- sapply(1:k.max,
13               function(k){kmeans(data, k, nstart=10)$tot.withinss})
14 plot(1:k.max, wss,
15      type="b", pch = 19, frame = FALSE,
16      xlab="Number of clusters K",
17
18
19
20
21
22
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84
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92
93
94
95
96
97
98
99
100

```

Console

```

~/
* 1 proposed 9 as the best number of clusters
* 2 proposed 10 as the best number of clusters

**** Conclusion ****

* According to the majority rule, the best number of clusters is 3

*****
>

```

Environment

Global Environment

Data

Variable	Description
data	50 obs. of 4 variables
gap_stat	List of 6
km.res	List of 9
nb	List of 4
res.nb	List of 4
USArrests	50 obs. of 4 variables

Files Plots Packages Help Viewer

Zoom Export

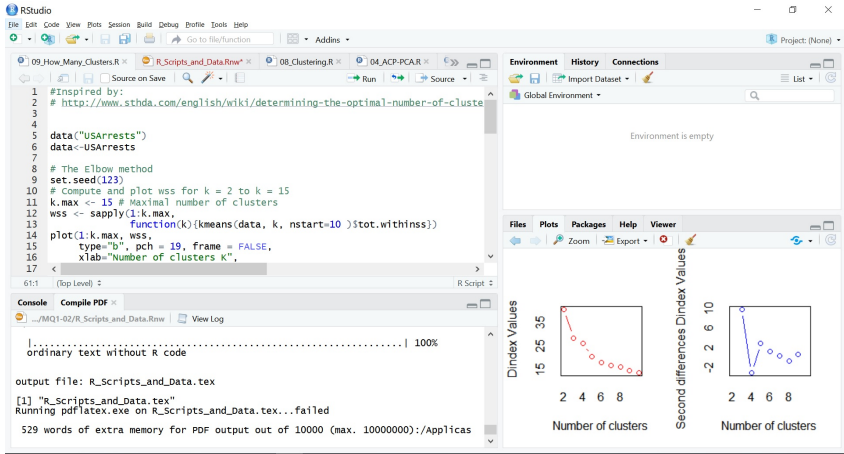
Index Values

Number of clusters

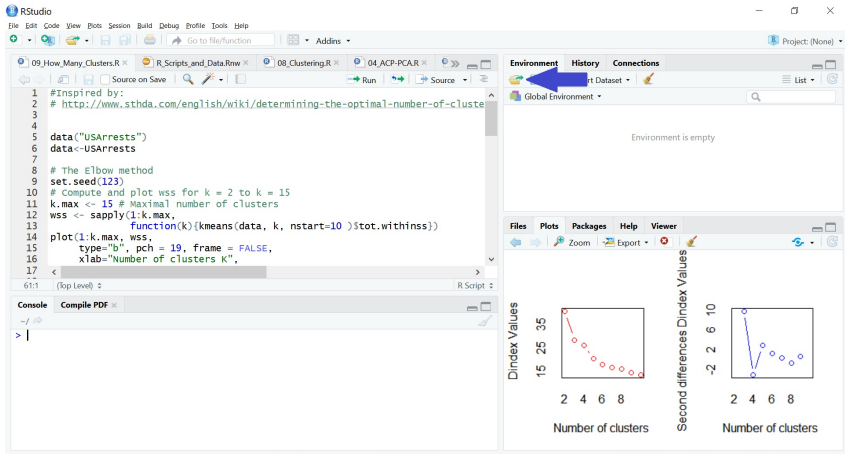
Second differences Index Values

Number of clusters

After

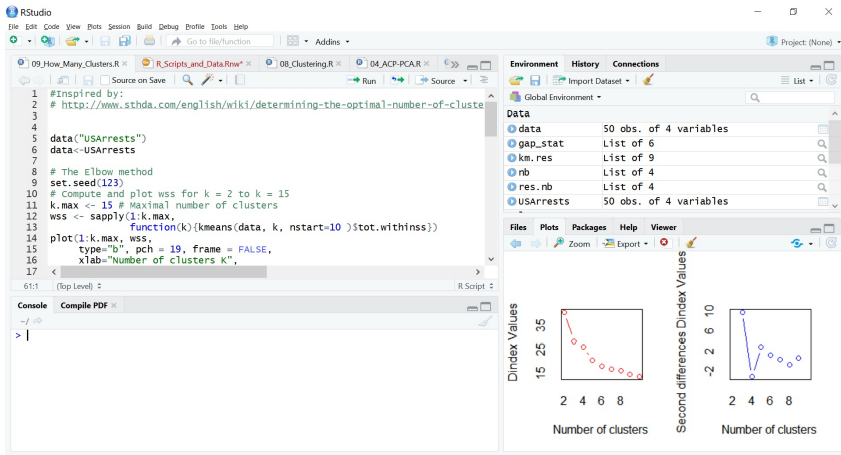


Before



File format is .RData

After



File format is .RData

- Packages are small toolboxes containing some functions to do some tasks.
- To use a package, you have to:
 - load it (one time) on your computer from CRAN (Comprehensive R Archive Network) using :

```
install.packages("clustree")
```

- load it into R everytime you need it using :

```
library(clustree)
```



- 1 Basics
- 2 Working with Scripts
- 3 Data Types and Data Structures
- 4 Statistics et probabilités
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- 6 Selecting Data and Variables
- 7 Managing Environment
- 8 **Statistics and Plotting for Multidimensional Data**



- Columns means :

```
colMeans (USArrests)
```

Murder	Assault	UrbanPop	Rape
7.788	170.760	65.540	21.232

- Covariance:

```
M<-cov (USArrests)
```

	Murder	Assault	UrbanPop	Rape
Murder	18.97	291.06	4.39	22.99
Assault	291.06	6945.17	312.28	519.27
UrbanPop	4.39	312.28	209.52	55.77
Rape	22.99	519.27	55.77	87.73

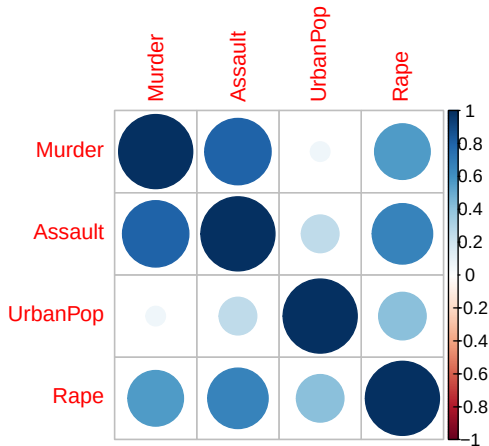
- Correlation:

```
C<-cor(USArrests)
```

	Murder	Assault	UrbanPop	Rape
Murder	1.00	0.80	0.07	0.56
Assault	0.80	1.00	0.26	0.67
UrbanPop	0.07	0.26	1.00	0.41
Rape	0.56	0.67	0.41	1.00

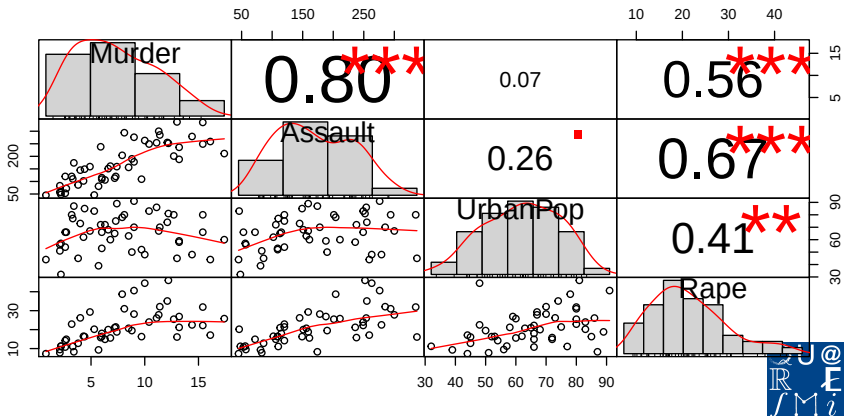
- Correlation:

```
library(corrplot)
corrplot(cor(USArrests))
```



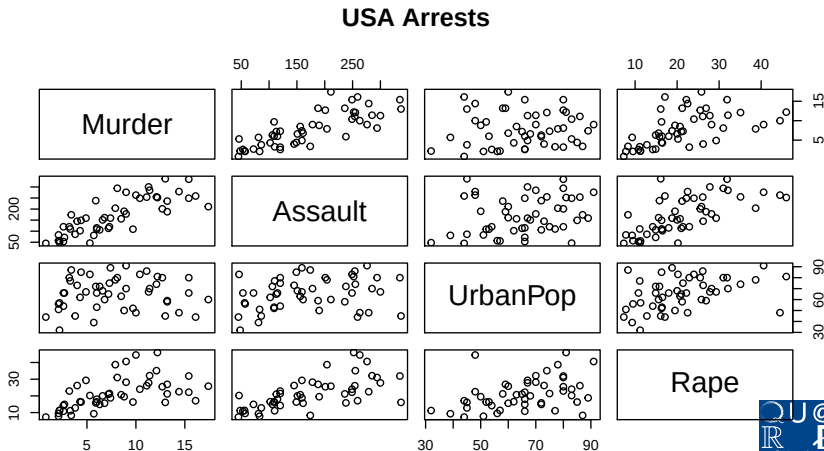
- Correlation:

```
library(PerformanceAnalytics)
chart.Correlation(USArrests, histogram=TRUE)
```



- Variables plotted by pairs:

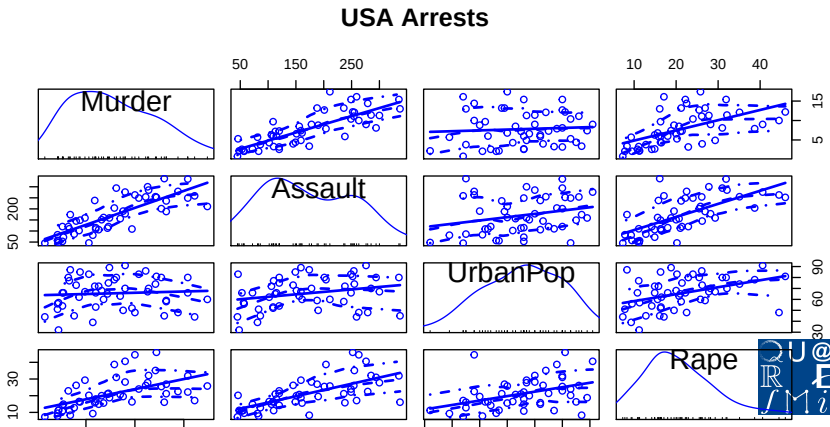
```
pairs(USArrests, main="USA Arrests")
```



- Variables plotted by pairs with regression lines:

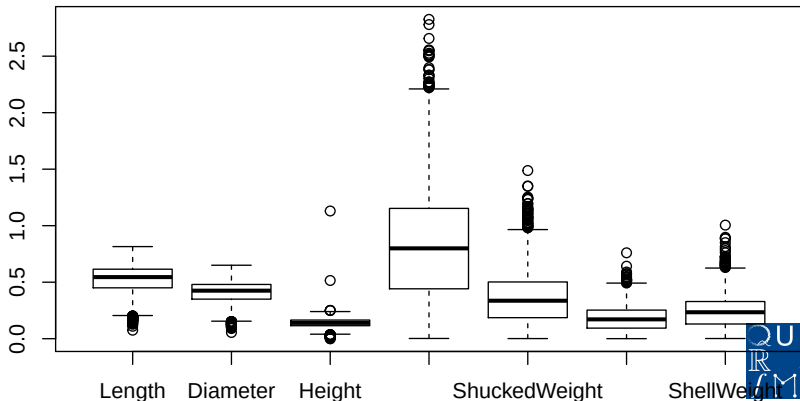
```
library(car)
```

```
scatterplotMatrix(USArrests, main="USA Arrests")
```



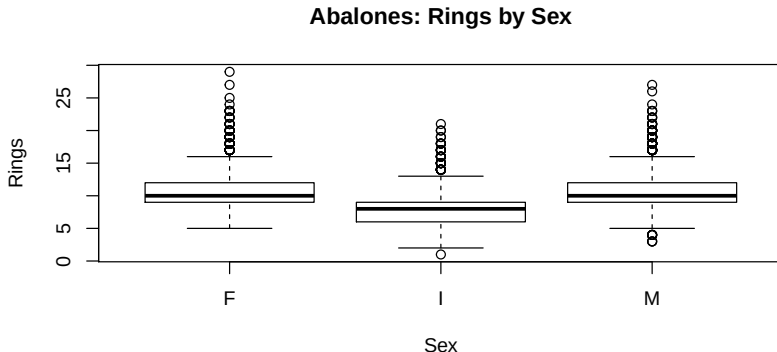
- Boxplot for several variables:

```
boxplot(abalone[, c(-1, -9)])
```



- Boxplot for one variable but several groups:

```
boxplot(Rings~Sex, data=abalone,  
main= "Abalones: Rings by Sex")
```



Note: "Rings~Sex" is called a formula in R and "means": "explain the variable Rings using the variable Sex".